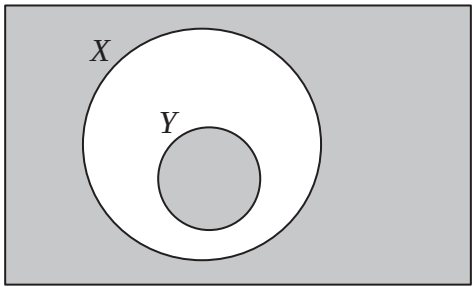


MATHEMATICS PAPER 2 ANSWER KEY

aQuestion		Solution
1	(a)(i)	As shown
	(a)(ii)	4
	(b)	$-\frac{1}{11a+7b}$
	(c)	5
2	(a)	\$462000
	(b)	\$62.50
	(c)	\$14499.92 (to nearest cent)
3	(a)	As shown
	(b)	5.86 cm
	(c)	3.77 cm ²
4	(a) (i)	$B = \{12, 15, 18, 21, 24, 27, 30\}$
	(a)(ii)	$n(A' \cap B') = 11$
	(a)(iii)	$A \cap B = C$
	(b)(i)	$\mathcal{E}$ 
	(b)(ii)	P'
5	(a)	12.6 cm (to 3 s.f)
	(b)	71.0° (to 1 dp)
	(c)	2.88 cm
6	(a)	$Q = \begin{pmatrix} 280 & 320 & 345 \\ 250 & 265 & 280 \\ 190 & 235 & 290 \end{pmatrix}$
	(b)(i)	$R = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$
	(b)(ii)	$\begin{pmatrix} 945 \\ 795 \\ 715 \end{pmatrix}$
	(c)	As shown
	(d)(i)	(2850.50)
	(d)(ii)	T represents the total amount of profit generated from the sales of AndClean hand sanitizers in all the 3 shops over the three months.

	(e)	$\begin{pmatrix} 483 \\ 392 \\ 406 \end{pmatrix}$
7	(a)	$p = -1.6$
	(c)	$x = 4.5$
	(d)	21
	(e)(ii)	$x = 2.9$
	(e)(iii)	$A = 10, B = -20$
8	(a)	35°
	(b)	290°
	(c)	15°
	(d)	105°
	(e)	30°
9	(a)(i)	The mean mark of class C was 75. Since the standard deviation of the class was zero, all students scored the same marks of 75.
	(a)(ii)	Class B had the best performance as the mean mark was the highest at 86.5. Class B also had the highest inconsistency in the scores at the standard deviation was the greatest at 2.128.
	(a)(iii)	$a = 84$ or $a = 66$
	(b)(i)	$\frac{7}{8}$
	b(ii)	$\left(\frac{7}{8}\right)^{n-1} \left(\frac{1}{8}\right)$
	b(iii)	$\left(\frac{7}{8}\right)^{n-1}$
	(c)	Not valid
10	(a)	1.53 cm
	(b)(i)	1.96 cm (to 3 sf)
	(b)(ii)	11.6% (to 3 sf)
	(c)(i)	6 cm
	(c)(ii)	5.20 cm (to 3 sf)
	(c)(iii)	9.46 cm
11	(a)	Defrosting required
	(b)	111 cm, 110 cm, 66 cm